SpaceX Falcon 9: can the first stage land successfully?

Business question. First-stage reuse influences launch economics. This project predicts landing success from historical mission characteristics; payload delivery success is a different target.

Executive summary. The 90-record modeling sample contains 60 successful landings (66.7%). Decision Tree wins training cross-validation (87.7%) and correctly predicts 15/18 holdout records (83.3%). Later flights generally show higher recovery success, but orbit, site and hardware are confounded.

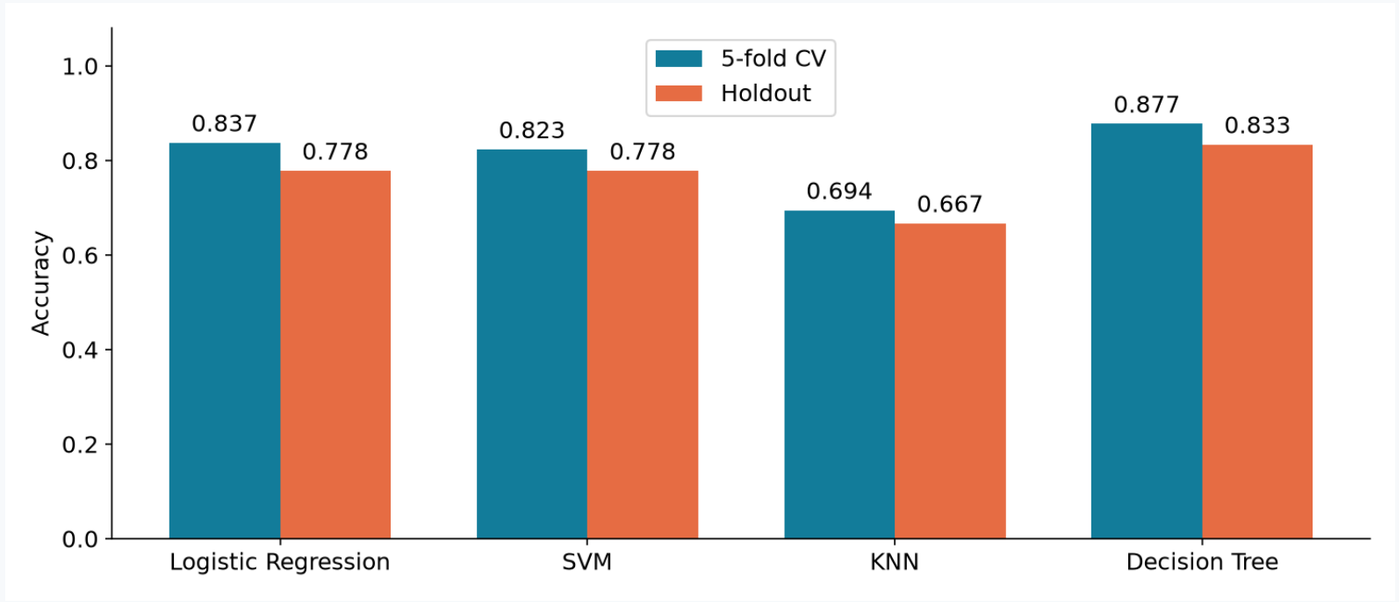
Public project repository: executed notebooks, Python files, datasets and results

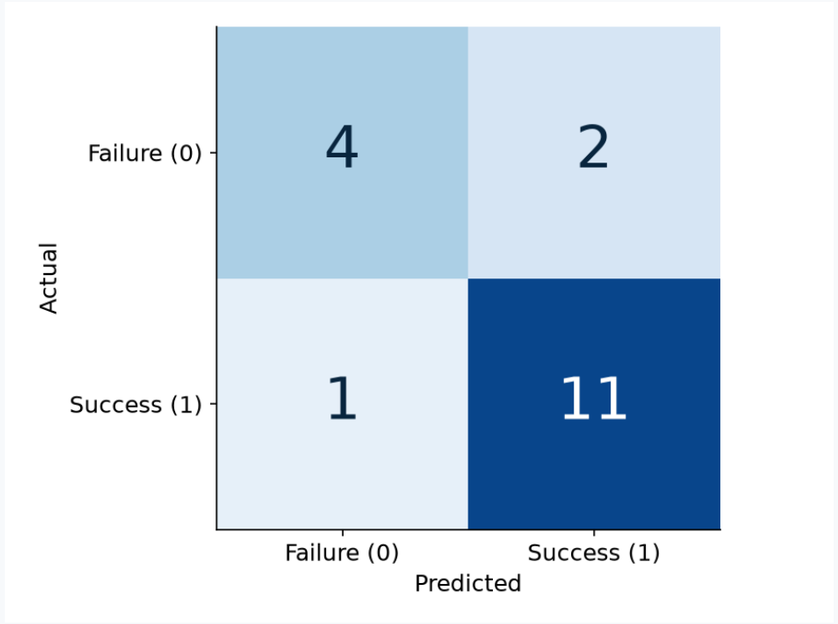
https://github.com/joaoltrevisan/spacex-falcon9-capstone

Evidence guide. Model results and confusion matrix: p. 2. Folium maps: pp. 3-4. Dash screenshots and payload interaction: pp. 5-6. Conclusions and original insights: p. 7. Methods and direct notebook URLs: pp. 8-10. Six EDA charts: pp. 11-13. Ten SQL queries and results: pp. 14-18.

Scope. Archived API: 107 records; historical scrape: 97 rows; model/EDA: 90 rows; SQL: 101 rows; Folium and Dash: 56 records each. These are separate laboratory samples, not one combined dataset.

Best classifier: Decision Tree; 15 of 18 holdout predictions correct





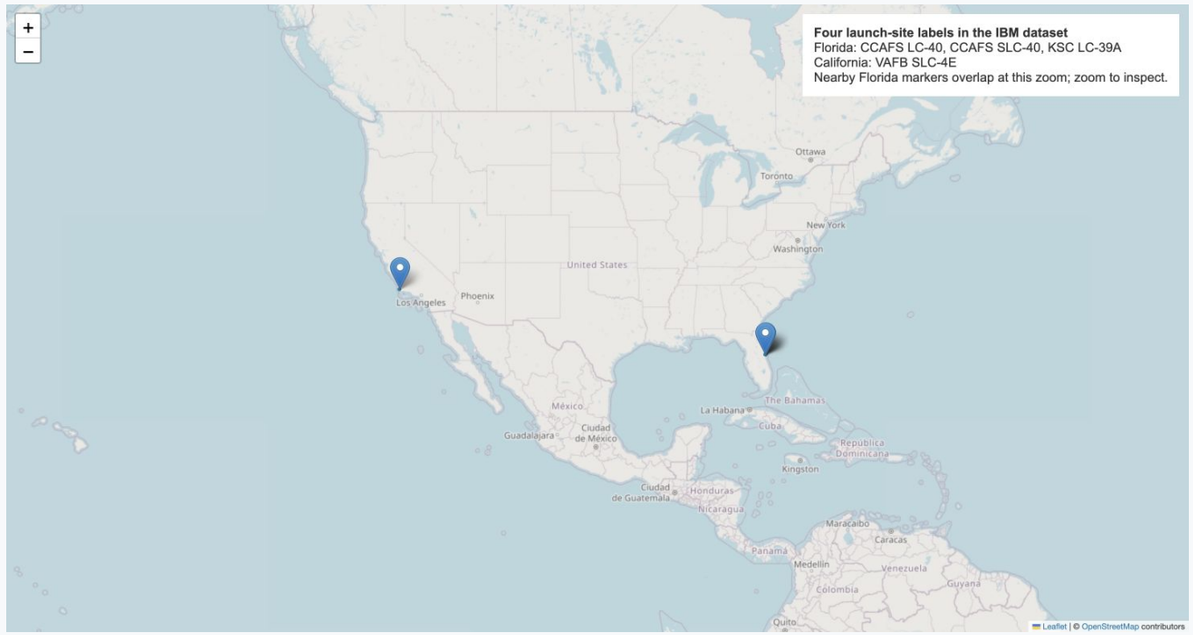
Selection: highest five-fold training CV accuracy, 0.877. Best tree parameters: gini, max_depth=5, min_samples_leaf=1. Holdout was excluded from tuning.

Confusion matrix: actual rows and predicted columns, ordered [failure, success]: **[[4, 2], [1, 11]]**. True negatives=4; false positives=2; false negatives=1; true positives=11.

Holdout metrics: accuracy 83.3%; precision 84.6%; recall 91.7%; F1 0.880. The majority baseline is 66.7%. One changed prediction shifts accuracy by 5.6 percentage points; temporal validation remains necessary.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

Folium: all sites and color-coded landing outcomes



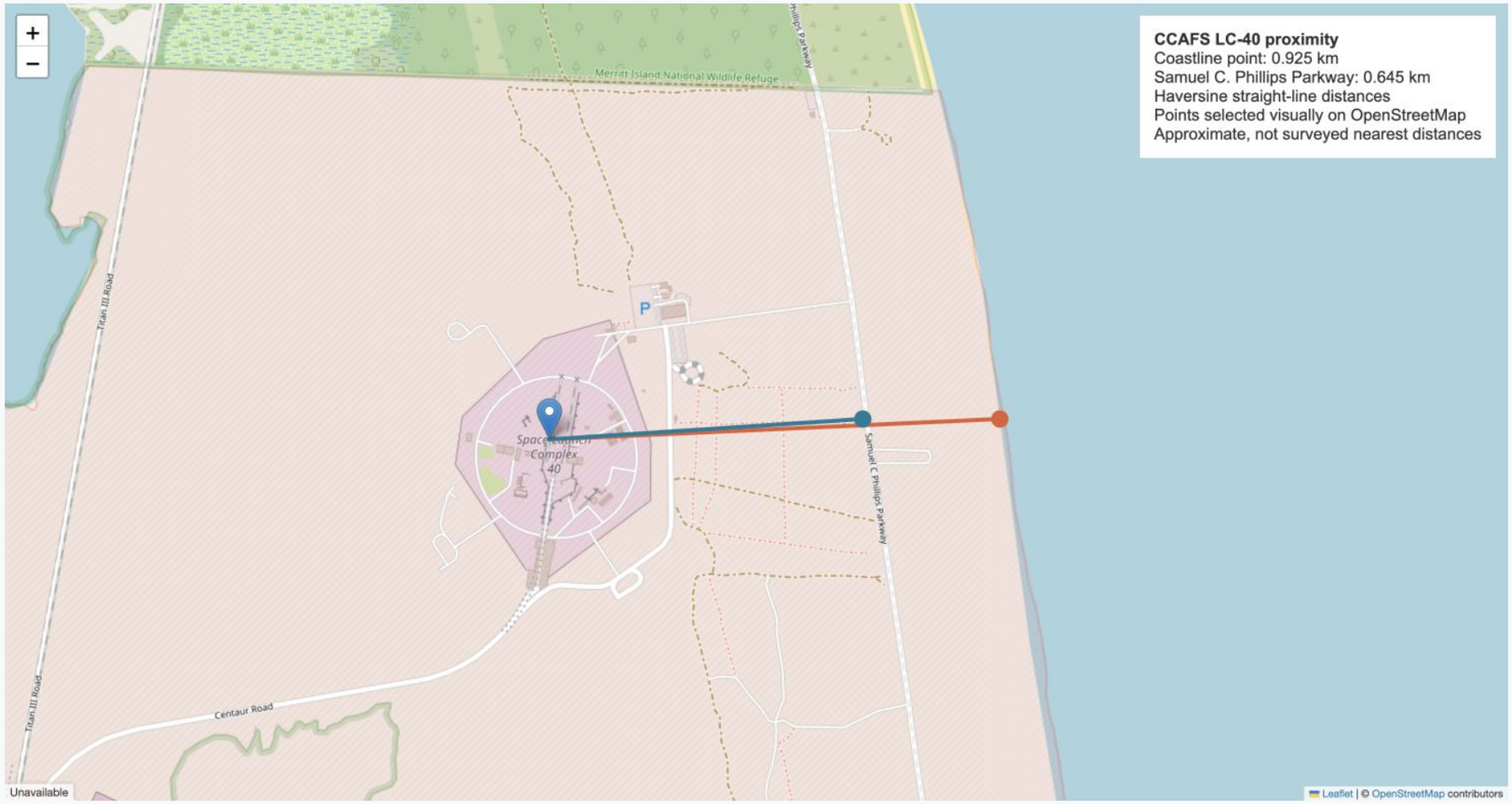


Left: national map of the four site labels in the 56-record geographic dataset: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E. Nearby Florida markers overlap at national zoom; the interactive map supports zoom and popups.

Right: expanded KSC launch cluster separates 13 overlapping records: 10 successes in green and 3 failures in red. The cluster preserves individual launch outcomes instead of showing only a site average.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Folium: measured proximity to coastline and roadway

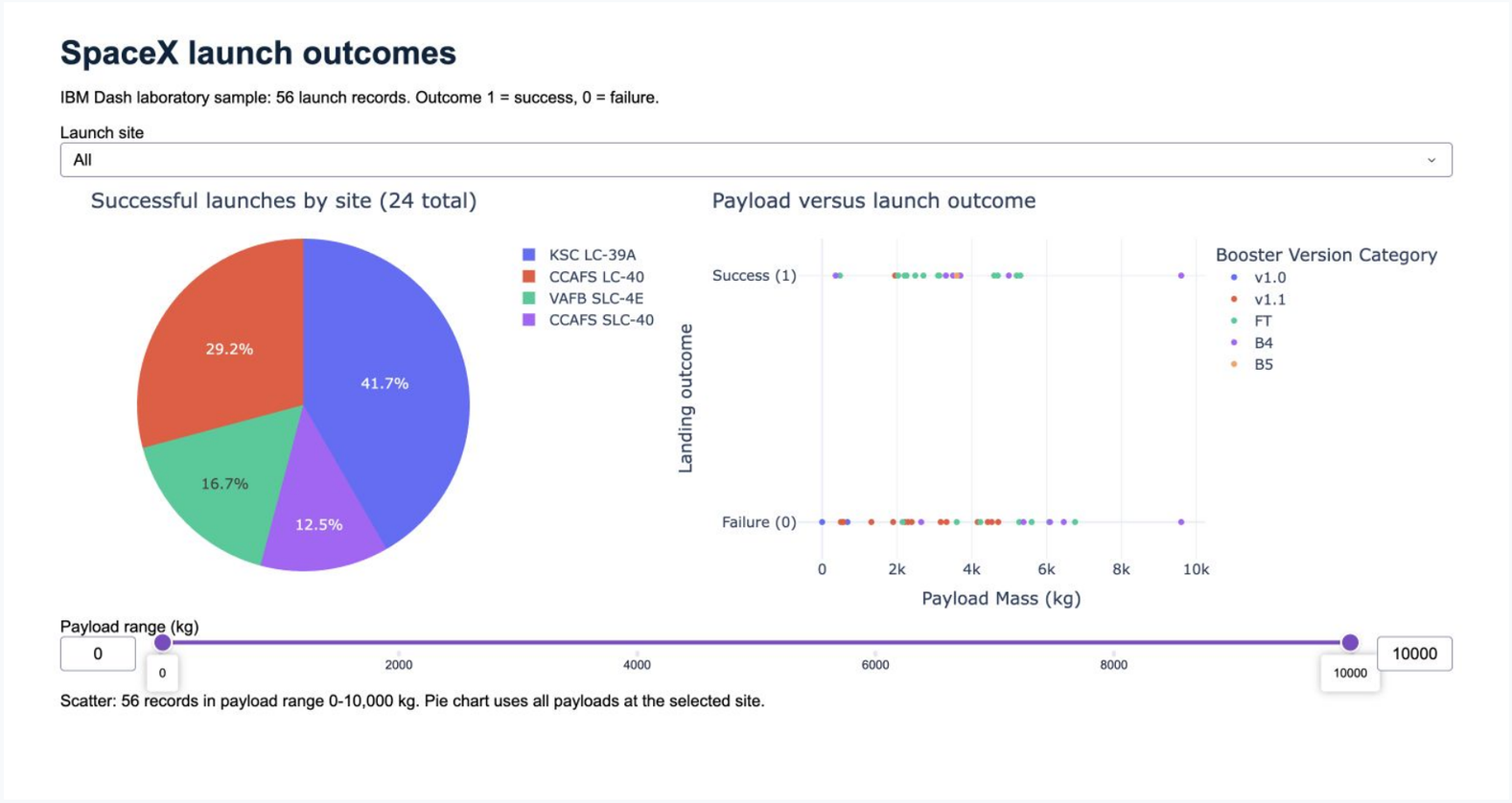


CCAFS LC-40: 0.925 km to coastline point (28.56268, -80.56789); 0.645 km to Samuel C. Phillips Parkway point (28.56268, -80.57077). Haversine distance uses the course launch-pad coordinates (28.56230197, -80.57735648).

Points were selected visually on OpenStreetMap using MousePosition. The colored lines show approximate straight-line proximity, not routes or surveyed nearest distances. Coastal location is observable; its causal effect on landing success is not established.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/src/build_folium_map.py

Dash: successful-launch counts and highest site success ratio

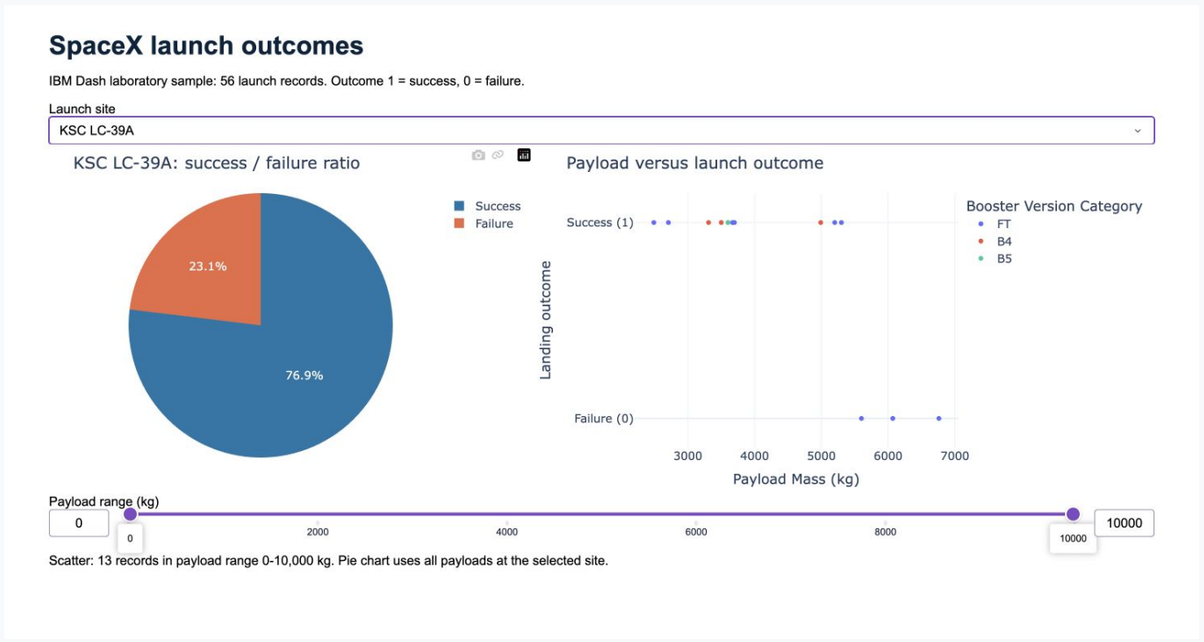


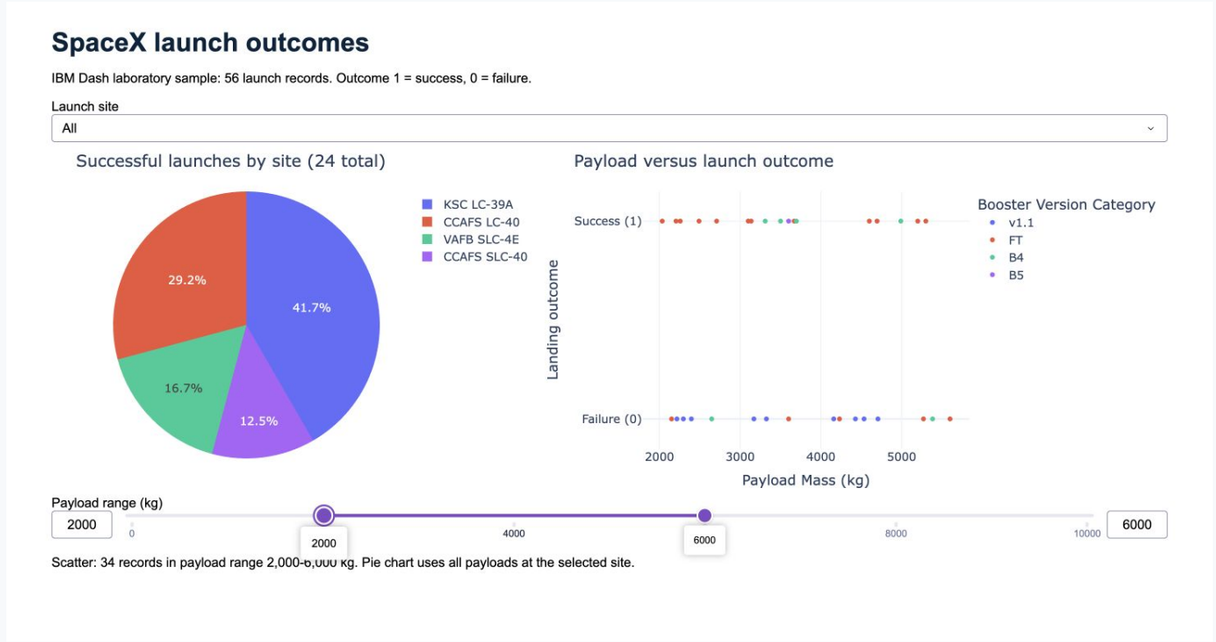
All-site pie: 24 successes in total: KSC 10 (41.7% of successes), CCAFS LC-40 7, VAFB 4, CCAFS SLC-40 3. These slices show success counts, not within-site success rates.

Within-site comparison: KSC 10/13=76.9%, CCAFS LC-40 7/26=26.9%, CCAFS SLC-40 3/7=42.9%, VAFB 4/10=40.0%. KSC has the highest rate in this sample. The selected-site view and slider comparison follow.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/06_Plotly_Dash_Dashboard.ipynb

Dash: site selection and payload RangeSlider in action





Left: dropdown set to KSC LC-39A; its pie displays 76.9% success (10/13). The scatter contains the 13 KSC records.

Right: All sites with payload range 2,000-6,000 kg: 34 records, 18 successes (52.9%). Full range 0-10,000 kg on p. 5 contains 56 records and 24 successes (42.9%). The slider changes the scatter sample; the pie keeps all payloads at the selected site.

Color identifies booster version. Overall FT succeeds in 16/24 launches, B4 in 6/11, B5 in 1/1, v1.0 in 0/5 and v1.1 in 1/15. B5's 100% is based on only one observation; it is not reliable evidence of superiority.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/src/dash_app.py

Conclusions and additional insights

• **Prediction:** Decision Tree improves on the holdout majority baseline by three correct predictions (15/18 versus 12/18). It identifies 11/12 successes and 4/6 failures; two failures are incorrectly predicted as successes.

• **Original insight - denominator sensitivity:** 90-row modeling baseline = 66.7%; 56-row dashboard baseline = 42.9%. The 23.8 percentage-point difference reflects sample coverage, not a measured operational improvement.

• **Original insight - selection changes the apparent rate:** filtering the dashboard to 2,000-6,000 kg changes observed success from 42.9% (24/56) to 52.9% (18/34). That association cannot establish a causal payload effect.

• **EDA conclusion:** later missions show more successes; SSO is 5/5 and GTO 14/27 in the modeling sample. Sparse orbits and correlated mission characteristics limit interpretation.

• **Next steps:** evaluate chronologically, remove post-launch information, test probability calibration and collect more independent observations before operational use.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

Collection and wrangling: reproducible sources and outcomes

• **API:** archived IBM JSON → validate list → flatten launch/first-core fields → sort by flight number → export 107 × 18 normalized table. This is archived-response processing; live API enrichment is not claimed.

• **Scraping:** fixed Wikipedia revision 987426997 → identify launch tables → clean headers/reference markers → parse UTC dates → filter 2010-06-04 to 2020-11-05 → deduplicate. Result: 97 rows.

• **Wrangling:** align the supplied 90 × 80 matrix with IBM labels by flight number and four one-hot category groups. Recreate encoding and confirm every feature value. Display missing landing-pad values as Unknown; retain the course's prepared numeric values.

• **Target:** Class 1 = successful landing (60 rows). Class 0 = other landing outcomes, including unsuccessful/unavailable outcomes (30 rows). Mission delivery success is separate.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/01_Data_Collection_API.ipynb

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/02_Data_Collection_Web_Scraping.ipynb

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/03_Data_Wrangling_and_Target.ipynb

EDA, SQL and interactive visualization methods

• **Charts:** four scatter plots compare flight/payload with site/orbit; color encodes Class. Orbit bars and annual lines show success rate with sample sizes. Purpose: inspect mission mix, chronology and sparse categories.

• **SQL:** trim text whitespace in the separate 101-row source; load SQLite; use DISTINCT, LIKE, grouped aggregates, MIN/MAX, a maximum-payload subquery and DENSE_RANK to answer ten lab questions.

• **Folium:** site circles and markers identify geography; MarkerCluster reveals overlapping outcomes; green/red icons encode success/failure; PolyLine and Haversine show coast/road proximity; MousePosition provides point coordinates.

• **Dash:** dropdown selects site; pie switches between success counts and within-site outcome ratio; scatter shows payload/outcome with booster color; RangeSlider filters payload and updates the sample count.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

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https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/src/dash_app.py

Classification methodology and tuned parameters

• **Split:** 90 observations; stratified training/holdout split of 72/18 with random_state=2. The holdout has 12 successes and 6 failures.

• **Tuning:** GridSearchCV with five-fold stratified training CV. Scaling is inside the Logistic Regression, SVM and KNN pipelines, avoiding scaling leakage across CV folds.

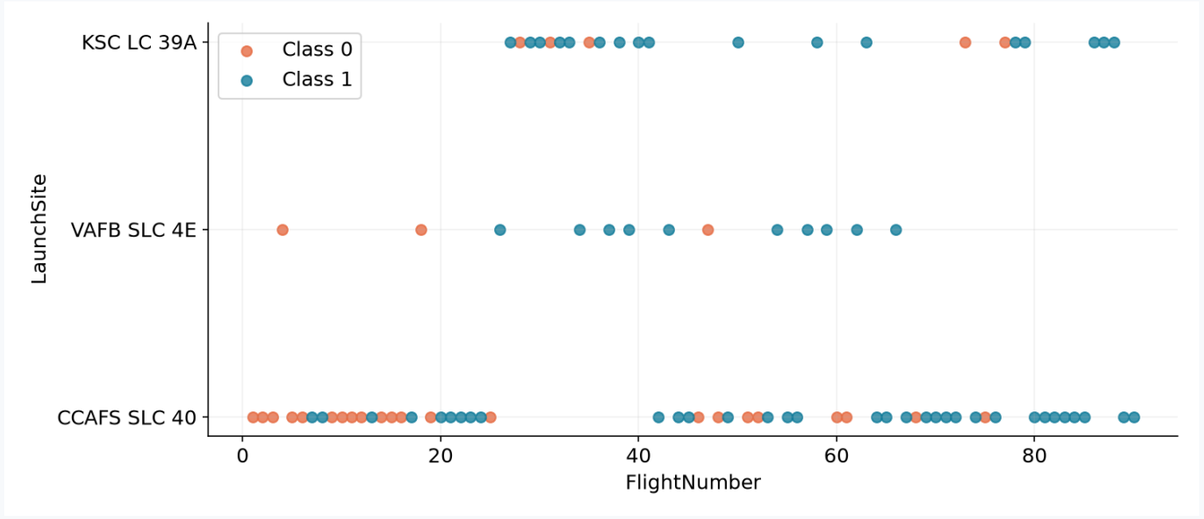
• **Models:** Logistic Regression (best C=1); SVM (linear, C=0.1); KNN (3 neighbors, uniform); Decision Tree (gini, depth 5, minimum leaf 1).

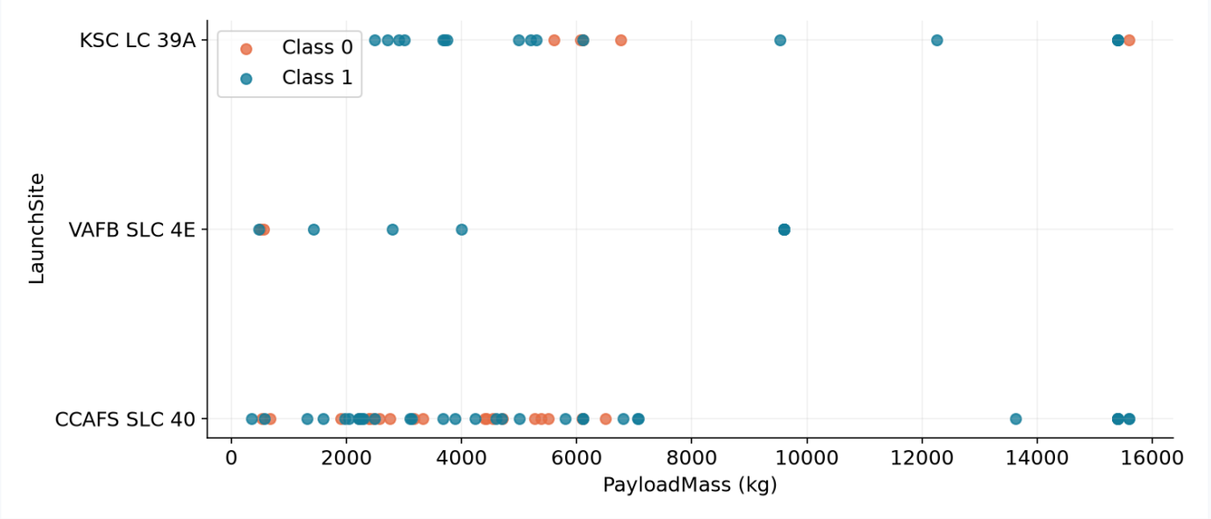
• **CV accuracy / holdout accuracy:** Logistic Regression 0.837/0.778; SVM 0.823/0.778; KNN 0.694/0.667; Decision Tree 0.877/0.833.

• **Interpretation:** select the tree by training CV, then inspect holdout confusion matrix and precision/recall (p. 2). KNN predicts every holdout row as success, matching the majority baseline. Random holdout performance is not evidence of future-launch forecasting accuracy.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

EDA: launch site, flight sequence and payload



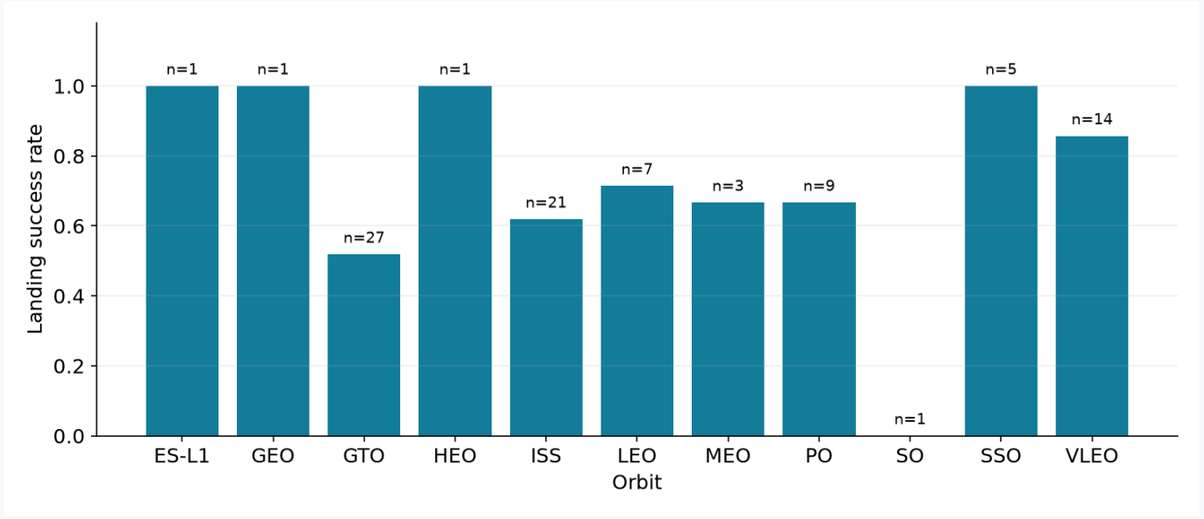


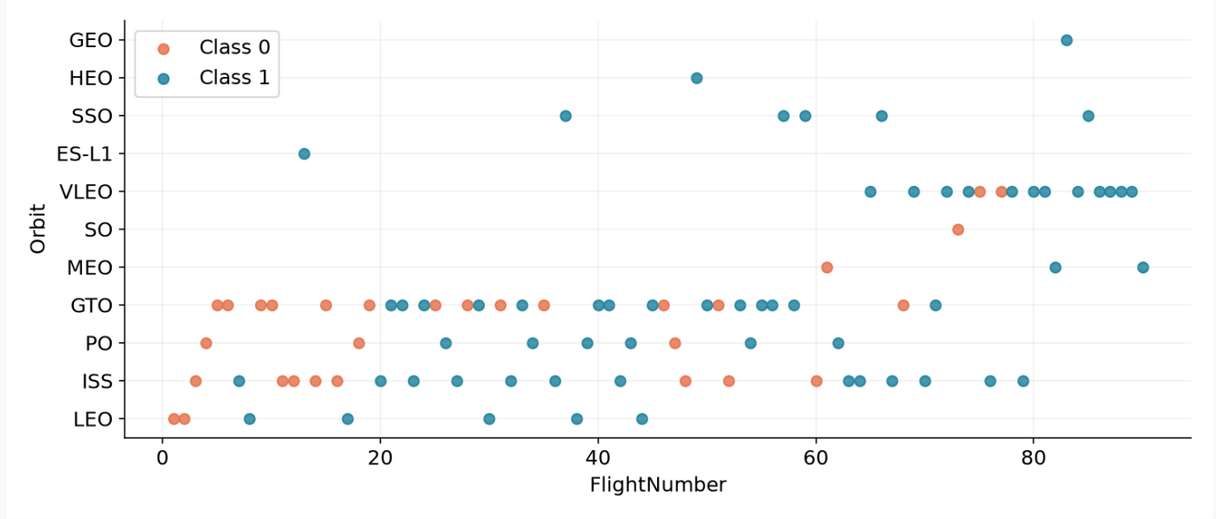
Left: Later flights show more successes across sites; flight sequence can proxy changes in experience and hardware.

Right: Site rates: KSC 17/22=77.3%, VAFB 10/13=76.9%, CCAFS 33/55=60.0%. Unequal sample sizes and mission mix matter.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

EDA: orbit success and mission sequence



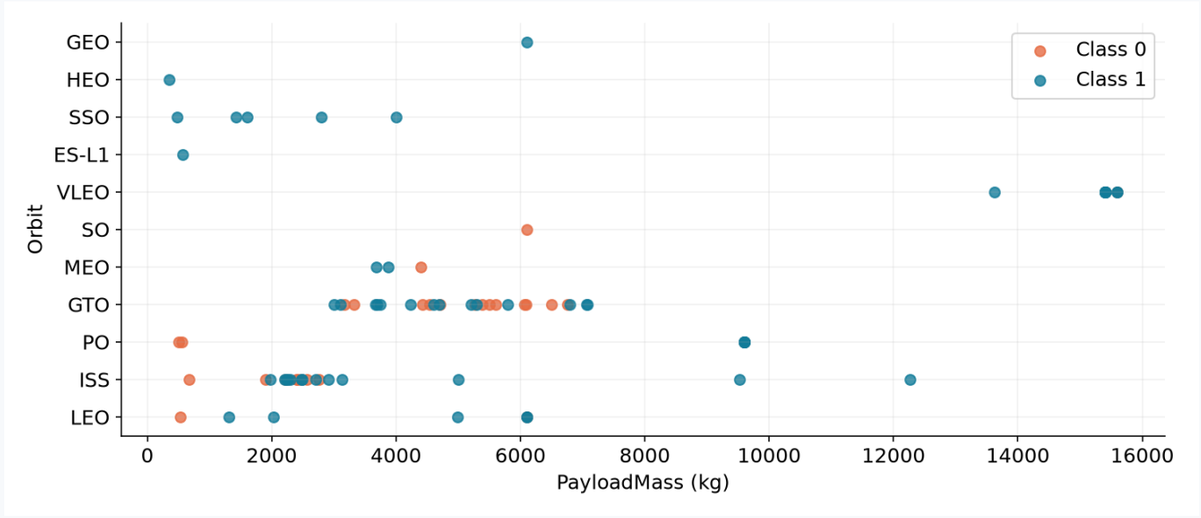


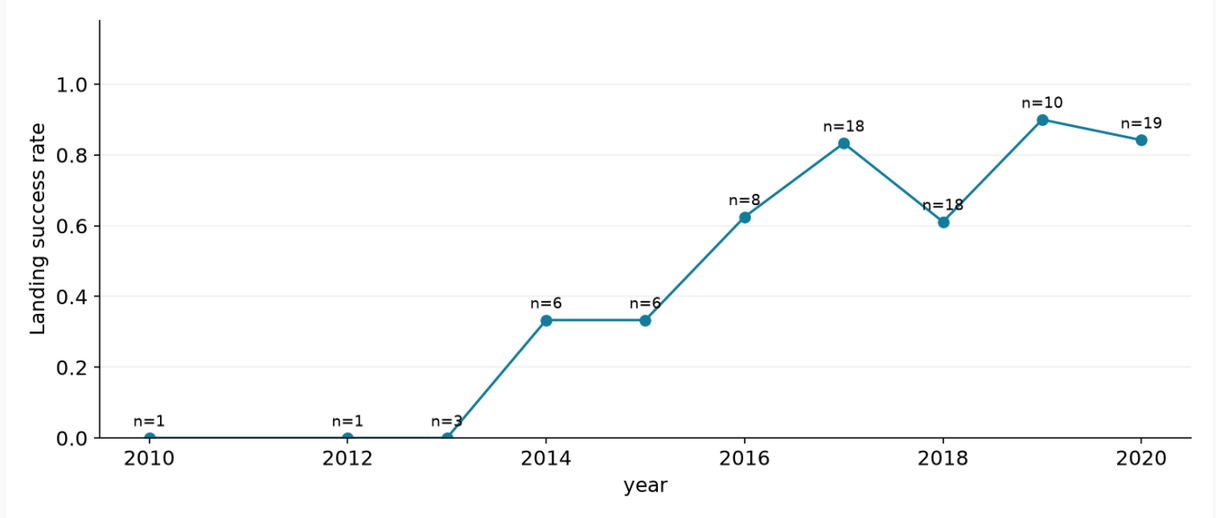
Left: SSO 5/5; VLEO 12/14; GTO 14/27. ES-L1, GEO and HEO each have only one record; their 100% rates are fragile.

Right: Orbit use changes over flight sequence. Descriptive differences cannot isolate the effect of orbit from chronology or hardware.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

EDA: payload by orbit and annual success





Left: Orbit and payload reflect mission requirements. Both outcomes occur across payloads; heavier payload is not proven to cause better recovery.

Right: 2017: 15/18; 2018: 11/18; 2019: 9/10; 2020: 16/19. The overall increase is not monotonic; early samples are small.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

SQL evidence | tasks 1 and 2

1. Unique launch sites

SELECT DISTINCT Launch_Site FROM launches ORDER BY Launch_Site

Launch_Site: CCAFS LC-40
Launch_Site: CCAFS SLC-40
Launch_Site: KSC LC-39A
Launch_Site: VAFB SLC-4E

Four distinct labels retained from the source.

2. First five chronological CCA records

SELECT Date, Booster_Version, Launch_Site, PAYLOAD_MASS__KG_ FROM launches WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5

Date: 2010-06-04 | Booster_Version: F9 v1.0 B0003 | Launch_Site: CCAFS LC-40 | PAYLOAD_MASS__KG_: 0
Date: 2010-12-08 | Booster_Version: F9 v1.0 B0004 | Launch_Site: CCAFS LC-40 | PAYLOAD_MASS__KG_: 0
Date: 2012-05-22 | Booster_Version: F9 v1.0 B0005 | Launch_Site: CCAFS LC-40 | PAYLOAD_MASS__KG_: 525
Date: 2012-10-08 | Booster_Version: F9 v1.0 B0006 | Launch_Site: CCAFS LC-40 | PAYLOAD_MASS__KG_: 500
Date: 2013-03-01 | Booster_Version: F9 v1.0 B0007 | Launch_Site: CCAFS LC-40 | PAYLOAD_MASS__KG_: 677

First five CCA records span 2010-2013.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

SQL evidence | tasks 3 and 4

3. Total NASA CRS payload

SELECT Customer, SUM(PAYLOAD_MASS__KG_) AS payload_kg FROM launches WHERE Customer = 'NASA (CRS)' GROUP BY Customer

Customer: NASA (CRS) | payload_kg: 45596

NASA (CRS) = 45,596 kg. All NASA mentions = 107,010 kg / 32 records.

4. Average F9 v1.1 payload

SELECT COUNT(*) AS launches, ROUND(AVG(PAYLOAD_MASS__KG_), 2) AS mean_payload_kg FROM launches WHERE Booster_Version LIKE 'F9 v1.1%'

launches: 15 | mean_payload_kg: 2534.67

F9 v1.1: 15 launches; mean payload 2,534.67 kg.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

SQL evidence | tasks 5 and 6

5. First ground-pad success

SELECT MIN(Date) AS first_successful_ground_landing FROM launches WHERE Landing_Outcome = 'Success (ground pad)'

first_successful_ground_landing: 2015-12-22

Earliest successful ground-pad landing: 2015-12-22.

6. Successful droneship, 4000 < payload < 6000 kg

SELECT Date, Booster_Version, PAYLOAD_MASS__KG_ FROM launches WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000 ORDER BY Date

Date: 2016-05-06 | Booster_Version: F9 FT B1022 | PAYLOAD_MASS__KG_: 4696
Date: 2016-08-14 | Booster_Version: F9 FT B1026 | PAYLOAD_MASS__KG_: 4600
Date: 2017-03-30 | Booster_Version: F9 FT B1021.2 | PAYLOAD_MASS__KG_: 5300
Date: 2017-10-11 | Booster_Version: F9 FT B1031.2 | PAYLOAD_MASS__KG_: 5200

Four records meet the successful landing and strict payload conditions.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

SQL evidence | tasks 7 and 8

7. Mission outcomes

SELECT Mission_Outcome, COUNT(*) AS missions FROM launches GROUP BY Mission_Outcome ORDER BY missions DESC

Mission_Outcome: Success | missions: 99
Mission_Outcome: Success (payload status unclear) | missions: 1
Mission_Outcome: Failure (in flight) | missions: 1

99 Success; 1 Success (payload status unclear); 1 Failure (in flight). These are not landing outcomes.

8. Boosters with maximum payload

SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM launches WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM launches) ORDER BY Date

F9 B5 B1048.4; F9 B5 B1049.4; F9 B5 B1051.3; F9 B5 B1056.4; F9 B5 B1048.5; F9 B5 B1051.4; F9 B5 B1049.5; F9 B5 B1060.2; F9 B5 B1058.3; F9 B5 B1051.6; F9 B5 B1060.3; F9 B5 B1049.7; each = 15,600 kg.

Maximum 15,600 kg; 12 booster records share that maximum.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

SQL evidence | tasks 9 and 10

9. 2015 failed droneship landings

SELECT Date, substr(Date,6,2) AS month, Booster_Version, Launch_Site, Landing_Outcome FROM launches WHERE Date LIKE '2015-%' AND Landing_Outcome = 'Failure (drone ship)' ORDER BY Date

Date: 2015-01-10 | month: 01 | Booster_Version: F9 v1.1 B1012 | Launch_Site: CCAFS LC-40 | Landing_Outcome: Failure (drone ship)
Date: 2015-04-14 | month: 04 | Booster_Version: F9 v1.1 B1015 | Launch_Site: CCAFS LC-40 | Landing_Outcome: Failure (drone ship)

Two failures: January 10 and April 14, both CCAFS LC-40.

10. Landing outcome ranking: 2010-06-04 to 2017-03-20

SELECT Landing_Outcome, COUNT(*) AS launches, DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rank FROM launches WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY launches DESC, Landing_Outcome

Landing_Outcome: No attempt | launches: 10 | rank: 1
Landing_Outcome: Failure (drone ship) | launches: 5 | rank: 2
Landing_Outcome: Success (drone ship) | launches: 5 | rank: 2
Landing_Outcome: Controlled (ocean) | launches: 3 | rank: 3
Landing_Outcome: Success (ground pad) | launches: 3 | rank: 3
Landing_Outcome: Failure (parachute) | launches: 2 | rank: 4
Landing_Outcome: Uncontrolled (ocean) | launches: 2 | rank: 4
Landing_Outcome: Precluded (drone ship) | launches: 1 | rank: 5

Inclusive date range; tied outcome counts share a dense rank.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb